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7.Write a program to implement Logistic Regression (LR) algorithm in python

1.

Study Hours	**Attendance**	**Result**
2	60	Fail
3	65	Fail
5	75	Pass
6	80	Pass
8	90	Pass
4	70	Fail

PROGRAM

```
import pandas as pd
from sklearn.linear_model import LogisticRegression
df = pd.read_csv("student.csv")
X = df[["Study Hours", "Attendance"]]
y = df["Result"]
model = LogisticRegression()
model.fit(X, y)
sample = [[7, 85]]
print("Prediction:", model.predict(sample)[0])
```

OUTPUT

Prediction: Pass

2.

Income	Credit Score	Loan			
3	580	No			
4	620	No			
6	700	Yes			
7	750	Yes			
8	780	Yes			
9	800	Yes			

PROGRAM

```
import pandas as pd
from sklearn.linear_model import LogisticRegression
df = pd.read_csv("loan.csv")
X = df[["Income", "Credit Score"]]
y = df["Loan"]
model = LogisticRegression()
model.fit(X, y)
sample = [[5, 650]]
print("Prediction:", model.predict(sample)[0])
```

OUTPUT

Prediction: No

3.

Temperature	Heart Rate	Disease
39.0	110	Positive
38.5	108	Positive
37.0	78	Negative
36.8	75	Negative
39.2	115	Positive

37.2	80	Negative

PROGRAM

```
import pandas as pd
from sklearn.linear_model import LogisticRegression

df = pd.read_csv("disease.csv")
X = df[["Temperature", "Heart Rate"]]
y = df["Disease"]
model = LogisticRegression()
model.fit(X, y)
sample = [[38.8, 112]]
print("Prediction:", model.predict(sample)[0])
```

OUTPUT

Prediction: Positive

4.

Experience	Performance	Promotion
2	60	No
3	65	No
4	70	No
6	80	Yes
7	85	Yes
9	95	Yes

PROGRAM

```
import pandas as pd
from sklearn.linear_model import LogisticRegression
df = pd.read_csv("employee.csv")
```

```
X = df[["Experience", "Performance"]]
y = df["Promotion"]
model = LogisticRegression()
model.fit(X, y)
sample = [[3, 68]]
print("Prediction:", model.predict(sample)[0])
```

OUTPUT

Prediction: No

5.

Weight	**Sweetness**	**Fruit**
120	70	Orange
125	72	Orange
130	74	Orange
145	90	Apple
150	92	Apple
160	88	Apple

PROGRAM

```
import pandas as pd
from sklearn.linear_model import LogisticRegression
df = pd.read_csv("fruit.csv")
X = df[["Weight", "Sweetness"]]
y = df["Fruit"]
model = LogisticRegression()
model.fit(X, y)
sample = [[128, 73]]
```

```
print("Prediction:", model.predict(sample)[0])
```

OUTPUT

Prediction: Orange